Reg. No. | | | | | | | | | | | [13AE71/12AE71]

SEVENTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, DECEMBER – 2016

AEROSPACE CONTROL SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Assume suitable missing data (if any)

1. a. Define: **04**

- **i.** Signal
- **ii.** Feedback
- **iii.** Block diagram

b. Solve the differential equation using Laplace transformation **10**

$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2 x = 0 \text{ , } x(0)=a, \ \dot{x}(0) = b$$

2. a. Given the open loop plant and the controller as G and G_c respectively derive for a unit feedback the closed loop transfer function of the system. **04**

b. For an L – R – C circuit derives the mathematical model and hence the plant transfer function. What is the order of the system? Also find the natural frequency and the damping ratio expressed in terms of the elements. Assume zero initial condition if required. **10**

3. a. Define for an under-damped second order system: **04**

- **i.** Settling time (t_s),
- **ii.** Rise time (t_r),
- **iii.** Peak time (t_p),
- **iv.** Maximum overshoot (Mp)

b. For a unity feedback control system with **10**

$$G(s) = \frac{64}{s(s+9.6)}$$

Write the output response to a unit step input. Determine:

- **i.** The response at t=0.1 sec.
- **ii.** Maximum value of the response and the time at which it occurs
- **iii.** Settling time.

4. a. Discuss analytically the problem for the proportional controller in a first order process. **04**

b. Explain analytically how the problem can be eliminated using the proportional integral (PI) controller. **05**

c. Show with an example the derivative control action adds damping to the system. **05**

5. a. Define and explain the significance of angle and magnitude condition, as applied to the root locus method of stability analysis of the linear system. **06**

b. For the following characteristic polynomial $s^2 + 2s + 2 + K(s+1) = 0$, draw the root locus for $0 \le K \le \infty$. **08**

6. Sketch the Bode plot for $G(s)H(s) = \frac{2}{s(s+1)(1+0.2s)}$. Also obtain gain margin phase margin and crossover frequencies. **14**

7. Consider a unity-feedback control system whose feed forward transfer function is given by **14**

$$G(s) = \frac{10}{s(s+2)(s+8)}$$

Design a compensator such that the dominant closed-loop poles are located at $s = -2 \pm j2\sqrt{3}$ and the static velocity error constant K_v is equal to 80 sec^{-1}.

8. a. Consider a two input-two output system: **07**

$$\frac{d^2y_1}{dt^2} + 2\frac{dy_1}{dt} + 3\frac{dy_2}{dt} + y_1 = u_1 + u_2 \text{ and}$$

$$\frac{d^2y_2}{dt^2} + 2\frac{dy_2}{dt} + 5\frac{dy_1}{dt} + y_2 = 2u_1 + 3u_2.$$

Derive a state model of the system.

b. Is the following system completely state controllable and completely observable? **07**

$$\begin{vmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{vmatrix} = \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ x_3 \end{vmatrix} + \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} u; \quad y = [20 \quad 9 \quad 1] \begin{vmatrix} x_1 \\ x_2 \\ x_3 \end{vmatrix}$$